ORCHESTRA AI FAULT-TOLERANT ARCHITECTURE SPECIFICATION 2026
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Author: Senior Quantum Systems Architecture Group

1. SYSTEM OVERVIEW AND FAULT-TOLERANCE THRESHOLDS
Topological surface codes enforce fault-tolerant quantum error correction via spatial parity checks.
The critical threshold requirement for logical break-even in 2026 hardware is a 2-qubit gate fidelity exceeding 99.27 percent (corresponding to an error rate below 0.73 percent).

2. LATTICE TOPOLOGY AND SYNDROME EXTRACTION
Syndrome extraction cycles are pipelined with dedicated ancilla readout resonators.
Rotated surface code lattices reduce physical qubit overhead by approximately 40 percent relative to s
Stabilizer measurements execute within 200 nanoseconds using high-speed microwave flux-bias pulse

3. VERIFIED SAFETY CONSTRAINTS AND DECODING LATENCY
Real-time decoding latency must remain strictly below 1 microsecond to prevent syndrome backlog.
Minimum code distance d=5 required for memory benchmarks; d=7 required for fault-tolerant magic st
Physical error rates exceeding 1.2 percent trigger automated circuit breaker isolation of malfunctioning